

ELLIOT KAPLAN

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EMPLOYMENT

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|---|----------------|
| Stockholm University | 2026 – Present |
| · Assistant professor | |
| University of Mons | 2025 – 2026 |
| · FNRS postdoctoral fellow | |
| · Visitor at the Hausdorff Center for Mathematics during October 2025 | |
| Max Planck Institute for Mathematics | 2024 – 2025 |
| · Postdoctoral fellow | |
| McMaster University | 2021 – 2024 |
| · NSF postdoctoral fellow | |
| · Postdoctoral fellow at the Fields Institute from July 2021 to June 2022 | |
| · Visitor at the Max Planck Institute for Mathematics during June 2023 | |

EDUCATION

- University of Illinois Urbana-Champaign**
- Doctorate in mathematics, May 2021
 - Dissertation: *Derivations on o-minimal fields*, advised by Lou van den Dries
- Ohio University**
- Bachelor of Science, Honors Tutorial College, May 2015
 - Mathematics major, philosophy minor, graduated summa cum laude
 - Undergraduate thesis: *Initial embeddings in the surreal number tree*, advised by Philip Ehrlich

RESEARCH INTERESTS

Model theory and its applications. Especially o-minimality, valued fields, and differential fields.

PUBLICATIONS

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|--|-------------------|
| 16. Decomposing the automorphism group of the surreal numbers | 2026 ⁺ |
| With Lothar Sebastian Krapp and Michele Serra
<i>Conference proceedings of DDG40</i> (to appear) ↗ 📄 | |
| 15. Constant power maps on Hardy fields and transseries | 2026 |
| <i>J. Lond. Math. Soc.</i> , vol. 114, no. 3, paper no. e70709 (2026) ↗ 📄 | |
| 14. Generic derivations, differential largeness, and NTP₂ | 2026 |
| With Christoph Kesting
<i>Ann. Pure Appl. Logic</i> , vol. 177, no. 6, paper no. 103730, 9p. (2026) ↗ 📄 | |

13. **Monotone T -convex T -differential fields** 2025
 With Nigel Pynn-Coates
Model Theory, vol. 4, no. 1, pp. 55–100 (2025) [↗](#) [📄](#)
12. **Hilbert polynomials for finitary matroids** 2024
 With Antongiulio Fornasiero
Pacific J. Math., vol. 333, no. 2, pp. 273–308 (2024) [↗](#) [📄](#)
11. **A dichotomy for T -convex fields with a monomial group** 2024
 With Christoph Kesting
MLQ Math. Log. Q., vol. 70, no. 1, pp. 99–110 (2024) [↗](#) [📄](#)
10. **Liouville closed H_T -fields** 2023
J. Algebra, vol. 628, pp. 265–327 (2023) [↗](#) [📄](#)
9. **T -convex T -differential fields and their immediate extensions** 2022
Pacific J. Math., vol. 320, no. 2, pp. 261–298 (2022) [↗](#) [📄](#)
8. **Surreal ordered exponential fields** 2021
 With Philip Ehrlich
J. Symb. Log., vol. 86, no. 3, pp. 1066–1115 (2021) [↗](#) [📄](#)
7. **Generic derivations on o-minimal structures** 2021
 With Antongiulio Fornasiero
J. Math. Log., vol. 21, no. 2, paper no. 2150007, 45p. (2021) [↗](#) [📄](#)
6. **Pairs of theories satisfying a Mordell-Lang condition** 2020
 With Alexi Block Gorman and Philipp Hieronymi
Fund. Math., vol. 251, no. 2, pp. 131–160 (2020) [↗](#) [📄](#)
5. **Distality for the asymptotic couple of the field of logarithmic transseries** 2020
 With Allen Gehret
Notre Dame J. Form. Log., vol. 61, no. 2, pp. 341–361 (2020) [↗](#) [📄](#)
4. **Continuous regular functions** 2020
 With Alexi Block Gorman, Philipp Hieronymi, Ruoyu Meng, Erik Walsberg, Zihe Wang, Ziqin Xiong, and Hongru Yang
Log. Methods Comput. Sci., vol. 16, no. 1, paper no. 17, 24p. (2020) [↗](#) [📄](#)
3. **Logarithmic hyperseries** 2019
 With Lou van den Dries and Joris van der Hoeven
Trans. Amer. Math. Soc., vol. 372, no. 7, pp. 5199–5241 (2019) [↗](#) [📄](#)
2. **Number systems with simplicity hierarchies II** 2018
 With Philip Ehrlich
J. Symb. Log., vol. 83, no. 2, pp. 617–633 (2018) [↗](#) [📄](#)
1. **Resolution depth of positive braids** 2015
 With David Krcatovich and Patricia O’Brien
J. Knot Theory Ramifications, vol. 24, no. 14, paper no. 1550075, 12p. (2015) [↗](#) [📄](#)

PREPRINTS

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4. **Building trees in large fields** 2026
 With Angus Matthews and Erik Walsberg
Submitted [↗](#) [📄](#)

3. **Geometric fields, ranks, and generic derivations** 2026
 With Antongiulio Fornasiero and Angus Matthews
Submitted  
2. **Dimension theory for the asymptotic couple of the field of logarithmic transseries** 2025
 With Allen Gehret and Nigel Pynn-Coates
Submitted  
1. **Hyperserial fields** 2021
 With Vincent Bagayoko and Joris van der Hoeven
Submitted  

AWARDS AND FELLOWSHIPS

F.R.S.-FNRS Postdoctoral Research Fellowship (CR – Chargé de Recherches)	2025 – 2026
NSF Mathematical Sciences Postdoctoral Research Fellowship (MSPRF)	2021 – 2024
List of Teachers Ranked as Excellent by Their Students	2015, 2017 – 2020
Kuo-Tsai Chen Prize in Mathematics	2019
Waldemar J., Barbara G., and Juliette Alexandra Trjitzinsky Fellowship	2019
Graduate College Conference Travel Award	2016
U.S. Junior Oberwolfach Fellows NSF Grant (DMS-1641185)	2016

TALKS AT CONFERENCES, WORKSHOPS, AND SPECIAL SESSIONS

Beijing Model Theory Conference Beijing, China	June 2026
Differential Algebra and Related Topics XIV Île d'Oléron, France	May 2026
BeNeLux Mathematical Congress Mathematical Logic Special Session.	April 2026
Workshop on Tame Geometry and Combinatorics University of Leeds, School of Maths	March 2026
INdAM Intensive Period at the University of Naples Mini-Workshop on Recent Developments in the Model Theory of Fields.	May 2025
Centre International de Rencontres Mathématiques Conference on Tame Geometry.	February 2025
AMS-UMI International Joint Meeting in Palermo Model Theory Special Session.	July 2024
CMS Summer Meeting at the University of Saskatchewan Mathematical Logic in Canada Special Session.	June 2024
North American meeting of the Association of Symbolic Logic at Iowa State Model Theory Special Session.	May 2024
AMS Sectional Meeting at the University of Wisconsin-Milwaukee Model Theory Special Session.	April 2024
Banff International Research Station Workshop on Exponential Fields.	February 2024

Banff International Research Station Workshop on Neostability.	February 2023
Oberwolfach Research Institute for Mathematics Mini-Workshop on Topological and Differential Expansions of o-minimal Structures.	November 2022
Oberwolfach Research Institute for Mathematics Mini-Workshop on Surreal Numbers, Surreal Analysis, Hahn Fields and Derivations.	December 2016
Joint Math Meetings in Seattle, WA AMS-ASL Special Session on Surreal Numbers.	January 2016

SEMINAR AND COLLOQUIUM TALKS

Model Theory Seminar at the Kurt Gödel Research Center	June 2026
PKU Model Theory Seminar	March 2026
Oberseminar Algebra und Geometrie at TU Dortmund	June 2025
Géométrie et Théorie des Modèles at the IHP	May 2025
Séminaire de Logique, Algèbre, Arithmétique et Géométrie at UMONS	April 2025
Logic Colloquium at the Kurt Gödel Research Center	April 2025
Oberseminar Logik at Universität Bonn	January 2025
AI Seminar at the Czech Technical University in Prague	January 2025
Théorie des Modèles et Groupes at IMJ-PRG	November 2024
Séminaire Général de Logique at IMJ-PRG	November 2024
Logic Seminar at The Ohio State University	April 2024
Logic Colloquium at the Kurt Gödel Research Center	March 2024
Waterloo Logic Seminar	January 2024
Oberseminar Modelltheorie, Geometrie und Gruppentheorie at Uni Münster	June 2023
Guest talk at HHU Düsseldorf	June 2023
Oberseminar Logik at Universität Bonn	June 2023
Leeds Logic Seminar	May 2023
Online Logic Seminar	March 2023
Purdue University Model Theory and Applications Seminar	March 2023
Logic Seminar at The Ohio State University	February 2023
Waterloo Logic Seminar	January 2023
Geometry and Model Theory Seminar at the Fields Institute	October 2022
Fields Postdoc Colloquium	April 2022
Geometry and Model Theory Seminar at the Fields Institute	January 2022
Kolchin Seminar in Differential Algebra	October 2021
Model Theory Seminar at the Fields Institute	August 2021
Topological and Differential Expansions of O-minimal Structures Seminar	February 2021

Manchester Logic Seminar	June 2020
Notre Dame Logic Seminar	June 2020
Séminaire de Logique Lyon-Paris	May 2020
Caltech-UCLA Logic Seminar	February 2020
Logic Seminar at The Ohio State University	November 2019
Graduate Model Theory Seminar at UCLA	January 2018

OUTREACH/GENERAL AUDIENCE TALKS

Graduate Colloquium at the University of Bonn	June 2025
An introduction to model theory for a graduate student audience.	
Casual Math Seminar at McMaster	March 2023
An introduction to combinatorial game theory for an undergraduate/graduate student audience.	
Undergraduate Friday Seminar at UIUC	March 2021
An introduction to Conway's surreal numbers for an undergraduate student audience.	
Summer Illinois Mathematics Camp at UIUC	July 2018
A discussion of the basics of knot theory for a high school audience.	

TEACHING AND MENTORSHIP

McMaster University	2022 – 2023
· Instructor for Linear Algebra I (Fall 2023). · Instructor for Engineering Mathematics I (Fall 2022).	
University of Illinois at Urbana-Champaign	2015 – 2020
· Discussion section leader for Calculus II (Fall 2015, Summer 2020), Calculus III (Fall 2016, Spring 2017, Fall 2018, Fall 2019), and Applied Linear Algebra (Spring 2018, Fall 2020). · Head teaching assistant for Elementary Linear Algebra (Fall 2017) and Applied Linear Algebra (Spring 2018, Fall 2020). · Grader for Mathematical Logic (Fall 2016, Fall 2017) and General Topology (Fall 2017).	
Preliminary Arizona Winter School (PAWS)	Fall 2022
<i>The Southwest Center for Arithmetic Geometry</i> Course assistant for Ronnie Nagloo's course <i>Introduction to model theory with applications</i> .	
Program for Interdisciplinary and Industrial Internships at Illinois	Summer 2019
<i>University of Illinois at Urbana-Champaign</i> Workshop leader for a three-day workshop on advanced linear algebra topics for graduate students.	
Illinois Geometry Lab	Summer 2019
Graduate Team Leader for a group of three undergraduate students. I designed and ran a research project on three-player combinatorial games.	
Illinois Geometry Lab	Spring 2018
Graduate Mentor for a group of four undergraduate students. I helped lead a research project studying functions recognizable by Büchi automata.	

PROFESSIONAL SERVICE AND OUTREACH

Co-organizer: Oberseminar Logik at Universität Bonn	2024 – 2025
Volunteer: OSU Math Department Booth at the COSI Science Festival	May 2024
Co-organizer: McMaster-Waterloo Model Theory Learning Seminar	Spring 2024
Co-organizer: ASL Special Session on Tame Geometry and Applications to Analysis <i>At the Joint Math Meetings in Boston, MA</i>	January 2023
Co-organizer: Graduate Student Conference in Logic XXII	March 2021
Organizer: Nonstandard Analysis Learning Seminar at UIUC	Spring 2021
Graduate student member: Undergraduate Affairs Committee at UIUC	2019 – 2020
Volunteer: Illinois Geometry Lab Booth at Illini Fest – Chicago	July 2019
Co-organizer: Graduate Student Conference in Logic XVIII	April 2017

OTHER WORK EXPERIENCE

Summer Intern <i>Wolfram Research</i>	Summer 2016
Summer Researcher at UIUC <i>Part of the Program for Interdisciplinary and Industrial Internships at Illinois</i>	Summer 2015
Undergraduate Researcher at Michigan State University <i>Part of the Summer Undergraduate Research Institute in Experimental Mathematics</i>	Summer 2014
CRLA Certified Mathematics Tutor with Ohio University Tutoring Services <i>Lead Mathematics Tutor from Fall 2014 through Spring 2015</i>	2012 – 2015